

# HRMS v2.0 Migration - TO DO List

Living document - updated as tasks complete. Source of truth for the migration's remaining work.

## Milestone status

|                    |                                                                    |
|--------------------|--------------------------------------------------------------------|
| Completed tasks    | 25 / 28                                                            |
| Current branch     | main @ 1542029                                                     |
| Next task          | Phase 5: final cutover to public, then Phase 6 DuckDB decommission |
| Follow-up progress | 5 / 6 done                                                         |
| Phase 0-1 progress | 3 / 3 done                                                         |
| Phase 2 progress   | 2 / 2 done                                                         |
| Phase 3a progress  | 3 / 3 done                                                         |
| Phase 3b progress  | 9 / 9 done                                                         |
| Phase 4 progress   | 3 / 3 done                                                         |
| Phase 5 progress   | 0 / 1 done                                                         |
| Phase 6 progress   | 0 / 1 done                                                         |

## TO DO / status by phase

### Phase 0-1 (3/3 done)

|      |                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------|
| DONE | <b>Freeze the v1.0 DuckDB schema; inventory tables for ETL</b><br>Schema freeze note in docs/MIGRATION.md (08f7e40)                      |
| DONE | <b>Build v2.0 target schema (db/postgres_schema.sql + Alembic baseline)</b><br>50-table public schema, CC-01..CC-16 documented (08f7e40) |
| DONE | <b>One-time DuckDB -&gt; PostgreSQL ETL with reconciliation</b><br>hrms DB seeded; counts reconciled (08f7e40)                           |

### Phase 2 (2/2 done)

|      |                                                                                                                                    |
|------|------------------------------------------------------------------------------------------------------------------------------------|
| DONE | <b>DuckDB-&gt;psycopg adapter (db_backend.py): translate/strftime/autocommit</b><br>App runs on PostgreSQL legacy schema (b9164f2) |
| DONE | <b>Unit + browser suites green on PostgreSQL (legacy schema preserved)</b><br>29/29 unit, 15/15 Playwright (b9164f2)               |

### Phase 3a (3/3 done)

|      |                                                                                             |
|------|---------------------------------------------------------------------------------------------|
| DONE | <b>Argon2id password hashing; legacy bcrypt re-hashed on login</b><br>security.py (98563ee) |
| DONE | <b>Global CSRF enforcement (fetch wrapper + csrf_token field + token API)</b><br>98563ee    |
| DONE | <b>Server-side Redis sessions (REDIS_URL opt-in) + login rate limiting</b><br>98563ee       |

### Phase 3b (9/9 done)

|      |                                                                                                                                                      |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| DONE | <b>CC-01 identity rule enforced: scripts/check_cc_rules.py + PG-gated test</b><br>46 identity + 4 natural keys, sequences ahead of data (acb5219)    |
| DONE | <b>Public-flip readiness probe (scripts/probe_public_flip.py)</b><br>GET + write-flow matrix against a throwaway public DB (acb5219)                 |
| DONE | <b>Boolean adapter compat: predicates, INSERT params, naive datetime round-trip</b><br>Inert on legacy (zero boolean cols); PG-gated tests (21480fe) |

|      |                                                                                                                                                                                                                                                                             |
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| DONE | <b>Write-flow probe + salary_structures seed data fix (CC-05)</b><br>11/11 core write flows green on public (21480fe)                                                                                                                                                       |
| DONE | <b>CC-09 transactional outbox (outbox.py + scheduler + admin endpoints)</b><br>Atomic business-write + event; backoff -> dead-letter (7e52f88)                                                                                                                              |
| DONE | <b>CC-07 idempotency: @idempotent decorator + idempotency_keys wired for keyed POST retries</b><br>Replay-without-duplicate, 409 on body reuse, claim released on failure; 6 unit tests green on every stack; probe replays on pure v2.0 JSONB                              |
| DONE | <b>Service-layer rewrite inc 1: expanded audit_log (actor/entity/entity_id/before/after/request_id, CC-13) + notifications.category (FR-NOT-03)</b><br>8ff66bc; +6 unit tests; DuckDB 46 green                                                                              |
| DONE | <b>Service-layer rewrite inc 2: shift_assignments replaces users.shift_start/shift_end (FR-ATT-17); init_db no longer mutates v2.0 public.users</b><br>get_shift/set_shift helpers reroute ~10 touch points; user CRUD + seed via set_shift; public probe 42/42 write flows |
| DONE | <b>Extend probe write section: forgot-password, payroll bank-file/TDS, ticket/ATS + verify against a clean public schema</b><br>Clean hrms_probe re-run: 94/94 GET + 42/42 write flows, including attendance, payroll, and lifecycle paths; all seed-time writes green      |

## Phase 4 (3/3 done)

|      |                                                                                                                                                                 |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DONE | <b>Attendance finalisation job (FR-JOB-01)</b><br>7 acceptance tests + nightly scheduler + regularization recompute; clean public probe 94/94 GET + 42/42 write |
| DONE | <b>Maker-checker payroll (FR-PAY-06)</b><br>Strict state machine + approval trail + Finance/Admin UI; clean public probe 94/94 GET + 42/42 write                |
| DONE | <b>Corrected ATS / onboarding / offboarding flows</b><br>FR-ATS/FR-ONB/FR-OFF acceptance suite + 94/42 public probe                                             |

## Phase 5 (0/1 done)

|                |                                                                                                                                                                                                                |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IN<br>PROGRESS | <b>Final cutover: flip APP_DB_SCHEMA to public, retire legacy</b><br>Disposable rehearsal passed ETL/preflight/94-42 probe/authenticated smoke/CC-01 sequence check; maintenance-window traffic switch pending |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Phase 6 (0/1 done)

|         |                                                                                |
|---------|--------------------------------------------------------------------------------|
| PENDING | <b>Decommission DuckDB runtime</b><br>Out of the ETL blueprint; tracked in SRS |
|---------|--------------------------------------------------------------------------------|

## Follow-up (5/6 done)

|         |                                                                                                                                                                                                                                                                        |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DONE    | <b>FR-USR archive/restore and session revocation</b><br>Hard delete replaced with retained archive/restore; admin UI and cross-backend tests updated                                                                                                                   |
| DONE    | <b>FR-USR directory contract (bounded pagination, sorting, validation)</b><br>per_page capped at 200, allow-listed sorting, EMP/email/role/departement validation, case-insensitive uniqueness, partial PUT with role-change audit, CSV import under the same contract |
| DONE    | <b>FR-USR-09 permission policy (policy.py matrix + permissions API/UI)</b><br>27-module role matrix, deny-beats-allow overrides, full-replace PUT with audit diff, anti-lockout guard, admin permissions modal                                                         |
| DONE    | <b>FR-USR-15 policy enforcement wiring (decorators + navbar)</b><br>Role gates narrowed by policy.can(); navbar derived from the same gate/module tags; department grant modelled; nav/API drift tests                                                                 |
| DONE    | <b>FR-USR-15 inline scope checks + audited pii_reveal</b><br>No handler branches on the session role copy; can_view_all is the CC-11 scope half; PII reveal route audited per cross-employee read                                                                      |
| PENDING | <b>FR-USR policy leave balances, background bulk import jobs, anonymisation</b><br>Next backend slices after the PII reveal                                                                                                                                            |

## Test & readiness gates

| Gate                                             | Stack                  | Current result                                                                                              |
|--------------------------------------------------|------------------------|-------------------------------------------------------------------------------------------------------------|
| Unit suite (tests/test_app.py)                   | DuckDB                 | 102 passed, 6 skipped (PG-gated compatibility/public tests)                                                 |
| Unit suite (tests/test_app.py)                   | PostgreSQL             | 107 passed, 1 skipped (public-only shift test)                                                              |
| Unit suite (tests/test_app.py)                   | PostgreSQL + Redis     | 107 passed, 1 skipped                                                                                       |
| Browser suite (tests/test_playwright.py)         | DuckDB                 | 18 passed                                                                                                   |
| Browser suite (tests/test_playwright.py)         | PostgreSQL             | 18 passed                                                                                                   |
| CC-01 rule checker (scripts/check_cc_rules.py)   | hrms (public)          | OK - 46 identity + 4 natural keys                                                                           |
| Public-flip probe (scripts/probe_public_flip.py) | hrms_probe (public)    | 94/94 GET + 44/44 write flows; lifecycle + permission paths included                                        |
| Cutover preflight (scripts/cutover_preflight.py) | hrms (public)          | Ready: head 0003, identity/sequence rules, required tables, and delta report generated                      |
| Disposable Phase 5 rehearsal                     | hrms_cutover_rehearsal | ETL Phase-1/2 + CC-05 cleanup + preflight + 94/42 probe + authenticated smoke + CC-01 sequence check passed |

## Update log (newest first)

|                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>2026-09-28</b> | FR-USR-15 scope + PII: no handler branches on the session role copy any more (test-enforced), the company-wide list split became policy.can_view_all (CC-11 scope), the dashboard variant follows sees_admin_surface(), and pii_reveal is now enforced and audited: GET /api/users//pii is the only cross-employee personal-data read and every such reveal writes a PII_REVEAL audit row. DuckDB 102 passed / 6 skipped; PostgreSQL 107 passed / 1 skipped; Playwright 18 passed; probe 94/94 GET + 44/44 write.                                                                                                                   |
| <b>2026-09-28</b> | FR-USR-15 enforcement wiring: the four role gates now narrow with policy.can() (the role/department check stays the outer gate, so an empty override table is a no-op), every gated view is mapped to a module, and the navbar is derived from the same gate/module tags so a link exists only when the route would admit the user. The HR-department grant is modelled explicitly. Drift removed (Finance no longer sees 403ing links, Super Admin sees the admin links) and /admin/leaves is now reachable. DuckDB 97 passed / 6 skipped; PostgreSQL 102 passed / 1 skipped; Playwright 17 passed; probe 94/94 GET + 44/44 write. |
| <b>2026-09-28</b> | FR-USR-09 permission policy implemented: policy.py owns the 27-module role matrix with deny-beats-allow per-user overrides, GET/PUT /api/users//permissions does a full replace with an audit diff and an anti-lockout guard, and the admin user list gained a permissions modal. Empty overrides reproduce the role defaults exactly, so no existing authorization outcome changes; decorator/navbar enforcement is the next slice. DuckDB 90 passed / 6 skipped; PostgreSQL 95 passed / 1 skipped; Playwright 17 passed; public probe 94/94 GET + 44/44 write.                                                                    |
| <b>2026-09-28</b> | FR-USR directory contract implemented: GET /api/users caps per_page at 200 and sorts through an allow-list (400 on invalid input), POST/PUT validate the employee ID, email, role, department and status with case-insensitive uniqueness, PUT became a true partial update with a before/after audit diff, and the CSV import now validates every row and reports skipped rows. Admin UI mirrors the contract; the public probe stays green.                                                                                                                                                                                       |
| <b>2026-09-25</b> | Phase 5 cutover rehearsal completed: frozen-source ETL accepted a missing post-v1.0 payroll_approvals table, reconciled Phase-1/Phase-2 data, applied CC-05 cleanup, stamped head, passed read-only preflight and 94/42 public probe, and kept identity sequences ahead after boot seeds and authenticated GET/write smoke. Traffic switch remains a maintenance-window operation.                                                                                                                                                                                                                                                  |
| <b>2026-09-25</b> | FR-ATS/FR-ONB/FR-OFF corrected lifecycle implemented: guarded ATS state machine, 100% offer splits, atomic accepted-offer conversion, signed pre-boarding tokens and real-file validation, five-step guarded onboarding, parallel offboarding with F&F; maker-checker, and IST LWD access revocation, plus lifecycle hardening (strict split precision, ETL preservation, session revocation, encrypted credential delivery, and document authorization). DuckDB 77 passed / 5 skipped; PostgreSQL 81 passed / 1 skipped; Playwright 16 passed; public probe 94/94 GET + 42/42 write.                                               |
| <b>2026-09-24</b> | FR-PAY-06 maker-checker payroll implemented: Draft → Submitted → Approved → Finalized, Finance/Admin authorization, self-approval rejection, payroll_approvals trail, adjustment-run reference, Finance UI access, and public probe coverage. DuckDB 64 passed / 5 skipped; PostgreSQL 68 passed / 1 skipped; probe 86/86 GET + 30/30 write.                                                                                                                                                                                                                                                                                        |

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| <b>2026-09-24</b> | FR-JOB-01 attendance finalisation implemented: nightly per-shift-date classification, employee weekly-off patterns, holiday/leave precedence, transactional attendance_days replacement, calendar output, and public identity-key probe coverage. DuckDB 64 passed / 5 skipped; PostgreSQL 68 passed / 1 skipped; probe 86/86 GET + 30/30 write. |
| <b>2026-09-24</b> | Service-layer rewrite inc 2/3 completed: shift_assignments, explicit v2.0 INSERTs, BOOLEAN UPDATE parameter coercion, and binary payroll bank-file export; the clean public-flip probe is fully green.                                                                                                                                           |
| <b>2026-09-23</b> | CC-07 idempotent writes done: @idempotent decorator on 9 POST routes, idempotency_keys DDL + hourly purge, 6 unit tests green (DuckDB 40 / PG 43 / PG+Redis 43), probe replays on pure v2.0 JSONB -> 86/86 GET + 12/12 write. Next: service-layer rewrite.                                                                                       |
| <b>2026-09-23</b> | TODO list created. Next task slated: CC-07 idempotency.                                                                                                                                                                                                                                                                                          |

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